Hints & Solution :-

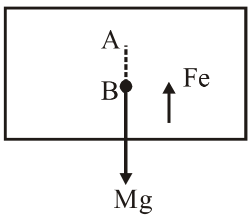
1. If body charged positively, mass will decrease & when negatively, mass will increase.

2. Additive Nature of charge define scalar nature of charge.

3. Charge is integral multiple of charge of electron is called quantization of charge.

4. $$a = \frac{(1/4\pi\in_0)e^2/r^2}{m_e} = 2.5\times10^{22}\ m/s^2$$

5. For, equilibrium Fe = Mg



For small vertical downward displacement $F_e' < F_e$ so it is unstable equilibrium

6. It will not be proper to say that number of field lines equal to EΔS cos θ

7. $\oint \vec{E}.\vec{ds} = \phi_{Enclosed}$

8. Valid When $F \propto 1/r^2$

9. The reason for these experience is a discharge of electric charges which are accumulated due to rubbing of insulating surfaces.

11. During its motion, body of carrier is charged due to rubbing with dry air and dust. If spark occurs near container, the inflammable material may catch fire. So, metallic ropes are suspended so, that excess charge flow away from carrier to ground.

15. Field is actually the property of source charge not of test charge.